

Final Project Guidelines

EEC 3150

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1 The Basics

The final project will give you a chance to flex your new Python skills in a manner of your choosing. You should try to utilize many of the tools we've learned throughout the course (don't just try to use as many as possible just for the sake of it though).

There are two parts to the final project:

- Code: you should have one or more .py files containing the code for your project
- Presentation: You will give a 10-15min presentation (w/ slides) about your project on Tuesday, May 2, 2:00pm-3:50pm (final exam time)

2 Expectations

2.1 Code

Your code should meet several expectations:

- Your code should be clearly readable (include some comments describing the different sections)
- Your code **SHOULD NOT HAVE ANY** bugs (at least for the intended use cases)
- You **MUST** use Matplotlib to visualize/plot some aspect of your project.

2.2 Presentation

Your presentation should have the following parts:

- Introduction: provide background material on the problem you're trying to solve with your project
- Methods: discuss the methods used to solve the problem
- Results: show your findings or give a demo of your project
- Conclusions: summarize your work and your key findings

Feel free to refer to your code throughout the presentation, especially for the results section.

3 Grading

Grading will be based on the following rubric:

Category	Max points
Code	100
- Neat and readable	(20)
- Matplotlib visualizations	(20)
- Effective solution, no bugs	(60)
Presentation	50
- All sections included	(20)
- Clear explanation of the project	(30)
Total	150

4 Example Project Ideas

Simple (relatively):

- Implement a game
 - Tic-tac-toe
 - Connect 4
 - Hangman
- Use an optimization method to solve an applied problem
- Data visualization and simple statistical analysis

Advanced:

- Simulate a simple differential equation with Euler's method
 - Exponential growth of a population
 - A planet orbiting a star
 - A pendulum
 - A particle in a box (w/ collisions)
- Regression, classification, or clustering on a dataset